

9. A rectangular coil of area A is kept in a uniform magnetic field $\vec{B}$ such that the plane of the coil makes an angle α with $\vec{B}$. The magnetic flux linked with the coil is :

(A) $BA \sin \alpha$

(B) $BA \cos \alpha$

(C) BA

(D) zero

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(C) BA

(D) zero

(A) $BA \sin \alpha$

9. A piece of a diamagnetic material, free to move when placed in a uniform magnetic field :

- (A) moves along the field
- (B) moves opposite to the field
- (C) moves perpendicular to the field
- (D) does not move at all

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4. A bar magnet is initially at right angles to a uniform magnetic field. The magnet is rotated till the torque acting on it becomes one-half of its initial value. The angle through which the bar magnet is rotated is :

(A) 30°

(B) 45°

(C) 60°

(D) 75°

4. A bar magnet is initially at right angles to a uniform magnetic field. The magnet is rotated till the torque acting on it becomes one-half of its initial value. The angle through which the bar magnet is rotated is :

(A) 30°

(B) 45°

(C) 60°

(D) 75°

(C) 60°

5. Which of the following substances has magnetic permeability less than that of free space ?

(A) Sodium

(B) Iron

(C) Aluminium

(D) Copper

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5. The materials having negative magnetic susceptibility are :

- | | |
|-------------------|------------------|
| (A) diamagnetic | (B) paramagnetic |
| (C) ferromagnetic | (D) non-magnetic |

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- | | |
|-------------------|------------------|
| (A) diamagnetic | (B) paramagnetic |
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(A) Diamagnetic

5. Which one out of the following materials is ***not*** paramagnetic ?

- (A) Aluminium
- (B) Sodium Chloride
- (C) Calcium
- (D) Copper Chloride

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(B) Sodium Chloride

4. A material is pushed out when placed in a uniform magnetic field. The material is

(A) non-magnetic

(B) diamagnetic

(C) paramagnetic

(D) ferromagnetic

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1

(A) non-magnetic

(B) diamagnetic

(C) paramagnetic

(D) ferromagnetic

(B) diamagnetic

4. A diamagnetic substance is brought, one by one, near the north pole and the south pole of a bar magnet. It is
- (A) repelled by north pole and attracted by south pole.
 - (B) attracted by north pole and repelled by south pole.
 - (C) attracted by north pole as well as by south pole.
 - (D) repelled by north pole as well as by south pole.

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- (A) repelled by north pole and attracted by south pole.
- (B) attracted by north pole and repelled by south pole.
- (C) attracted by north pole as well as by south pole.
- (D) repelled by north pole as well as by south pole.

(D) repelled by north pole as well as by south pole

4. The magnetic field due to a small magnetic dipole of dipole moment 'M' at a distance 'r' from the centre along the axis of the dipole is given by **1**

(A) $\frac{\mu_0}{4\pi} \times \frac{2M}{r^3}$

(B) $\frac{\mu_0}{4\pi} \times \frac{M}{r^3}$

(C) $\frac{\mu_0}{4\pi} \times \frac{M}{2r^3}$

(D) $\frac{\mu_0}{4\pi} \times \frac{2M}{r^2}$

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(D) $\frac{\mu_0}{4\pi} \times \frac{2M}{r^2}$

(A) $\frac{\mu_0}{4\pi} \times \frac{2M}{r^3}$

3. Which of the following material has its magnetic susceptibility χ in the range $0 < \chi < \epsilon$, where ϵ is positive and small ?

(A) Aluminium

(B) Water

(C) Gadolinium

(D) Bismuth

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(B) Water

(C) Gadolinium

(D) Bismuth

(A) Aluminum

3. An iron needle is kept near a strong bar magnet. It will experience
- (A) a force of attraction and no torque.
 - (B) a force of attraction and a torque.
 - (C) a torque and no force.
 - (D) neither a force nor a torque.

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(B) A force of attraction and a torque

3. Which of the following pairs is that of paramagnetic materials ?
- | | |
|--------------------------|------------------------|
| (A) Copper and Aluminium | (B) Sodium and Calcium |
| (C) Lead and Iron | (D) Nickel and Cobalt |

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| (C) Lead and Iron | (D) Nickel and Cobalt |

(B) Sodium and Calcium

11. The magnetic susceptibility for a diamagnetic material is **1**
- (A) small and negative (B) small and positive
- (C) large and negative (D) large and positive

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- (A) small and negative (B) small and positive
(C) large and negative (D) large and positive

(A) Small and negative.

- (b) What are ferromagnetic materials ? Explain ferromagnetism with the help of suitable diagrams, using the concept of magnetic domain. **3**

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(b)	Defining ferromagnetic materials	1
	Explanation of ferromagnetism with diagram	2

Ferromagnetic substances are those which get strongly magnetised when placed in an external magnetic field.



(a)



(b)

In absence of external magnetic field, domains are randomly oriented and it exhibits weak magnetisation.

In the presence of external magnetic field domains orient themselves in the direction of magnetic field and it exhibits strong magnetisation.

1

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

4. Which of the following is a diamagnetic substance ?

(A) Gadolinium

(B) Sodium

(C) Copper chloride

(D) Sodium chloride

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(B) Sodium

(C) Copper chloride

(D) Sodium chloride

(D) Sodium Chloride

5. A current carrying circular loop of magnetic moment $\vec{M}$ is suspended in a vertical plane in an external magnetic field $\vec{B}$ such that its plane is normal to $\vec{B}$. The work done in rotating this loop by 45° about an axis perpendicular to $\vec{B}$ is closest to :

(A) $-0.3 MB$

(B) $0.3 MB$

(C) $-1.7 MB$

(D) $1.7 MB$

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- | | |
|---------------|--------------|
| (A) $-0.3 MB$ | (B) $0.3 MB$ |
| (C) $-1.7 MB$ | (D) $1.7 MB$ |

(B) 0.3 MB

4. A circular coil of radius 10 cm is placed in a magnetic field $\vec{B} = (1.0 \hat{i} + 0.5 \hat{j})$ mT such that the outward unit vector normal to the surface of the coil is $(0.6 \hat{i} + 0.8 \hat{j})$. The magnetic flux linked with the coil is :

(A) $0.314 \mu\text{Wb}$

(B) $3.14 \mu\text{Wb}$

(C) $31.4 \mu\text{Wb}$

(D) $1.256 \mu\text{Wb}$

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(D) $1.256 \mu\text{Wb}$

(C) $31.4 \mu\text{Wb}$

5. When a ferromagnetic substance is heated to a temperature above its Curie temperature, it will :
- (A) behave like a diamagnetic material.
 - (B) behave like a paramagnetic material.
 - (C) permanently demagnetise.
 - (D) remain a ferromagnetic.

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- (A) behave like a diamagnetic material.
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 - (C) permanently demagnetise.
 - (D) remain a ferromagnetic.

(B) Behave like a paramagnetic material

1. The relative magnetic permeability of a medium is 0.075. Its magnetic susceptibility will be :

- (A) 0.925 (B) -0.925 (C) 1.075 (D) -1.075

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- (A) 0.925 (B) -0.925 (C) 1.075 (D) -1.075

(B) -0.925

18. Differentiate between magnetisation and the susceptibility of a material. What can you say about the susceptibility of paramagnetic and diamagnetic materials ?

2

18. Differentiate between magnetisation and the susceptibility of a material. What can you say about the susceptibility of paramagnetic and diamagnetic materials ?

2

Magnetization is equal to the net magnetic moment per unit volume.

Alternatively

$$\vec{M} = \frac{\vec{m}_{\text{net}}}{V}$$

Susceptibility is a measure of how a magnetic material responds to an external field.

Alternatively

$$\vec{M} = \chi \vec{H}$$

Susceptibility of paramagnetic material is between 0 & ϵ (where ϵ has small positive value)

Alternatively

$$0 < \chi < 1$$

Susceptibility of diamagnetic material is between $-1 \leq \chi < 0$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

5. Which of the following substances has relative magnetic permeability $\mu_r \gg 1$?

(A) Aluminium

(B) Copper chloride

(C) Nickel

(D) Sodium chloride

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(C) Nickel

(D) Sodium chloride

(C) Nickel

3. A bar magnet of magnetic moment 4 JT^{-1} is kept along the direction of magnetic field of 0.4 T . The work done to rotate the magnet through an angle of 90° is :

(A) Zero

(B) -1.6 J

(C) -0.2 J

(D) 1.6 J

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4. The dimensions of permeability of a medium are :

(A) $[MLT^{-1} A^{-1}]$

(B) $[MLT^{-2} A^{-2}]$

(C) $[ML^{-2} T^{-2} A^{-1}]$

(D) $[MLT^{-2} A^{-1}]$

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(B) $[MLT^{-2} A^{-2}]$

(C) $[ML^{-2} T^{-2} A^{-1}]$

(D) $[MLT^{-2} A^{-1}]$

(B) $[MLT^{-2} A^{-2}]$

6. Which one of the following has relative magnetic permeability between 0 and 1 ?

(a) Aluminium

(b) Alnico

(c) Water

(d) Sodium

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(a) Aluminium

(b) Alnico

(c) Water

(d) Sodium

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8. Which one of the following has negative value of susceptibility ?

(a) Nickel

(b) Aluminium

(c) Lead

(d) Iron

8. Which one of the following has negative value of susceptibility ?

(a) Nickel

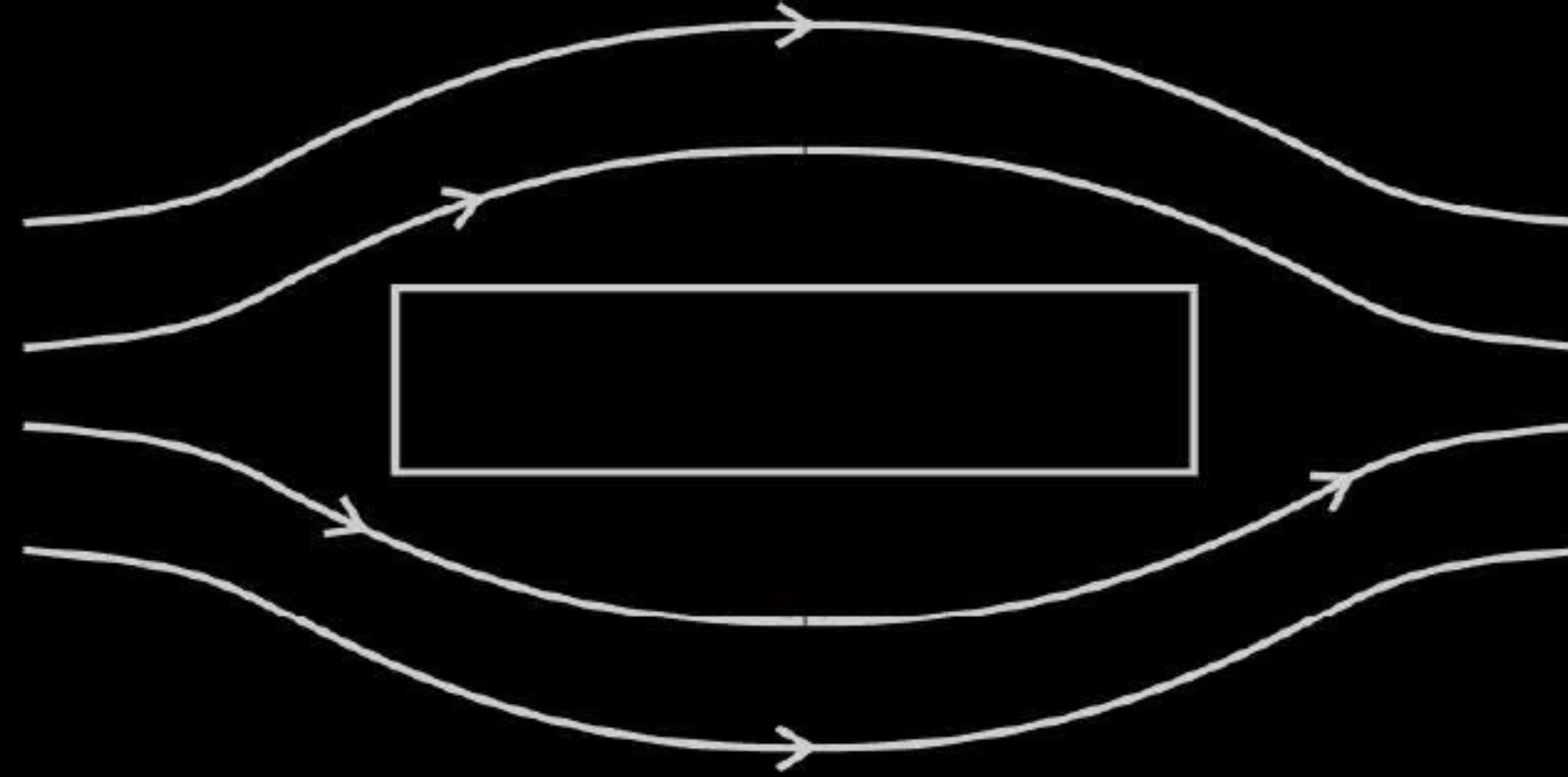
(b) Aluminium

(c) Lead

(d) Iron

(c) Lead

3. The magnetic field lines near a substance are as shown in the figure. The substance is :



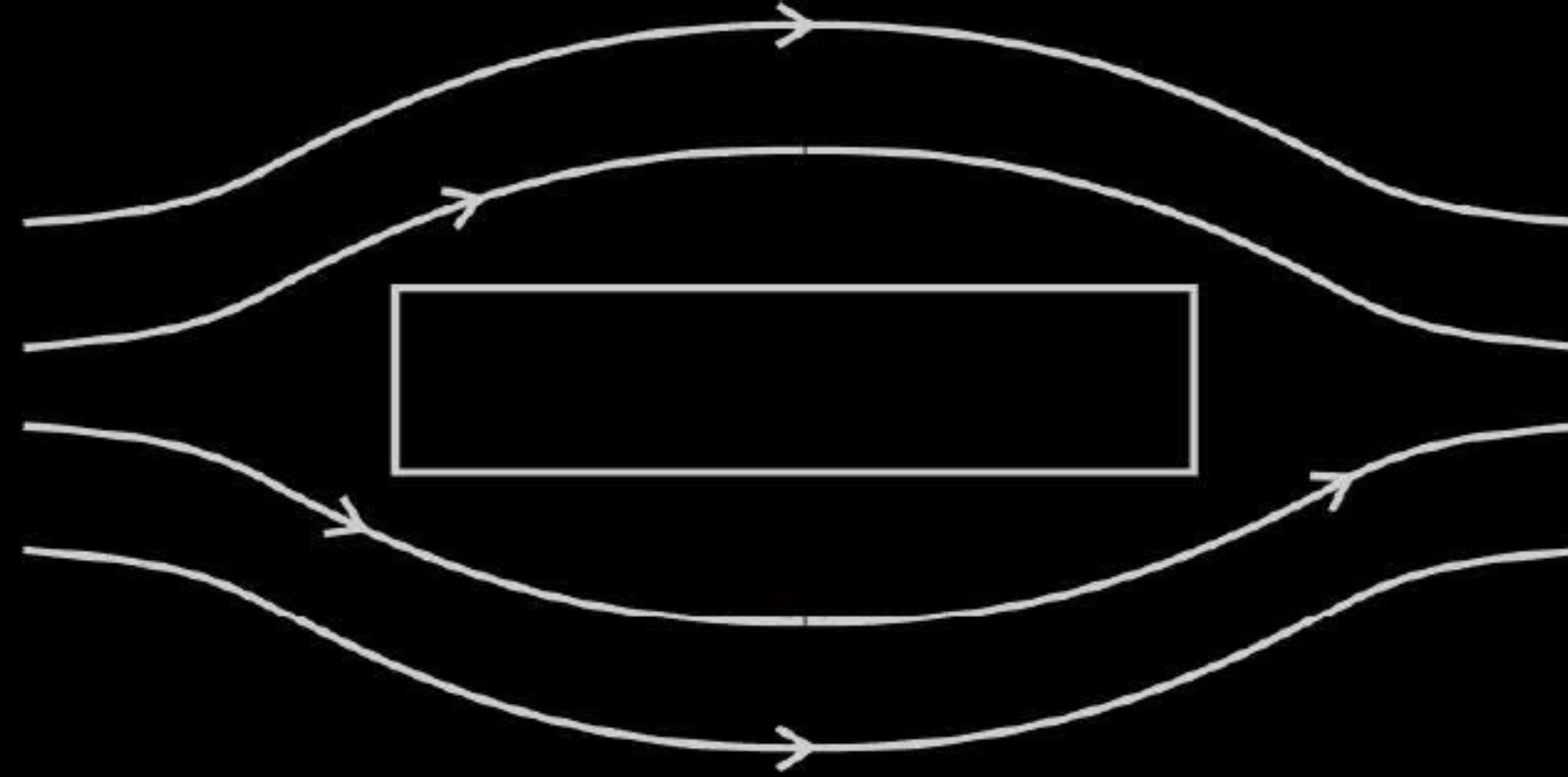
(a) Copper

(b) Iron

(c) Sodium

(d) Aluminium

3. The magnetic field lines near a substance are as shown in the figure. The substance is :



(a) Copper

(b) Iron

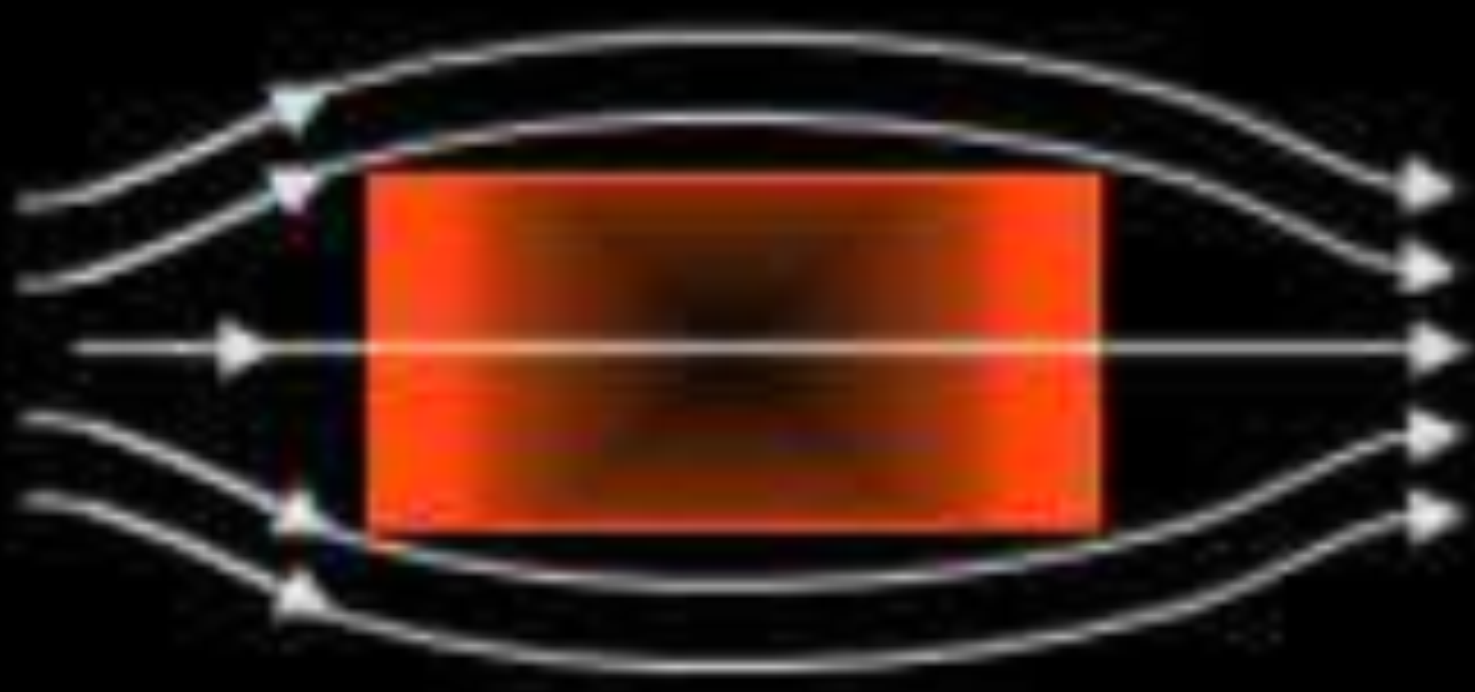
(c) Sodium

(d) Aluminium

(a) Copper

- (ii) Write any two points of difference between a diamagnetic and a paramagnetic substance.

(ii) Write any two points of difference between a diamagnetic and a paramagnetic substance.

Diamagnetic Materials (i) Susceptibility is between -1 and 0. (ii) Relative permeability is between 0 and 1. (iii) $\mu < \mu_0$ (iv) Tendency to move from stronger to weaker part of external magnetism. (v) is repelled by a magnet. (vi) Field inside the material is reduced. (vii) 	Paramagnetic Materials (i) Susceptibility is a small positive number.(slightly greater than zero.) (ii) Relative permeability is slightly greater than 1. (iii) $\mu > \mu_0$ (iv) Tendency to move from region of weak to strong magnetic field. (v) is weakly attracted by a magnet. (vi) Field inside is slightly enhanced. (vii) 	1+1
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Any two of the above mentioned differences.

12. A diamagnetic substance is brought near the north or south pole of a bar magnet. It will be :

- (a) repelled by both the poles.
- (b) attracted by both the poles.
- (c) repelled by the north pole and attracted by the south pole.
- (d) attracted by the north pole and repelled by the south pole.

(a) Repelled by both the poles.

17. *Assertion (A) :* When a bar of copper is placed in an external magnetic field, the field lines get concentrated inside the bar.

Reason (R) : Copper is a paramagnetic substance.

17. *Assertion (A) :* When a bar of copper is placed in an external magnetic field, the field lines get concentrated inside the bar.

Reason (R) : Copper is a paramagnetic substance.

(d) Assertion (A) is false and Reason (R) is also false.

17. **Assertion (A)** : Diamagnetic substances exhibit magnetism.

Reason (R) : Diamagnetic materials do not have permanent magnetic dipole moment.

17. **Assertion (A)** : Diamagnetic substances exhibit magnetism.

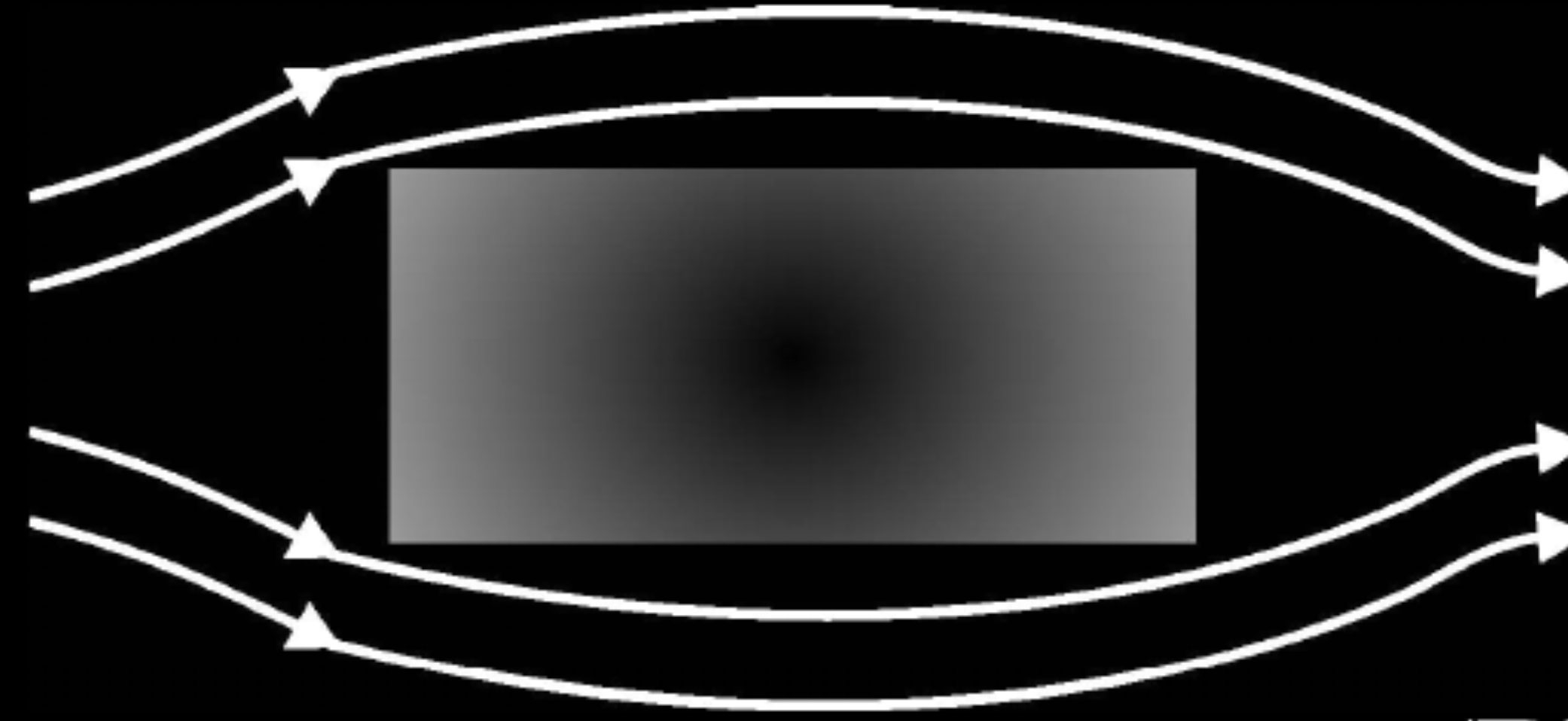
Reason (R) : Diamagnetic materials do not have permanent magnetic dipole moment.

1

(b) Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

3. Which of the following cannot modify an external magnetic field as shown in the figure ?

1



(A) Nickel

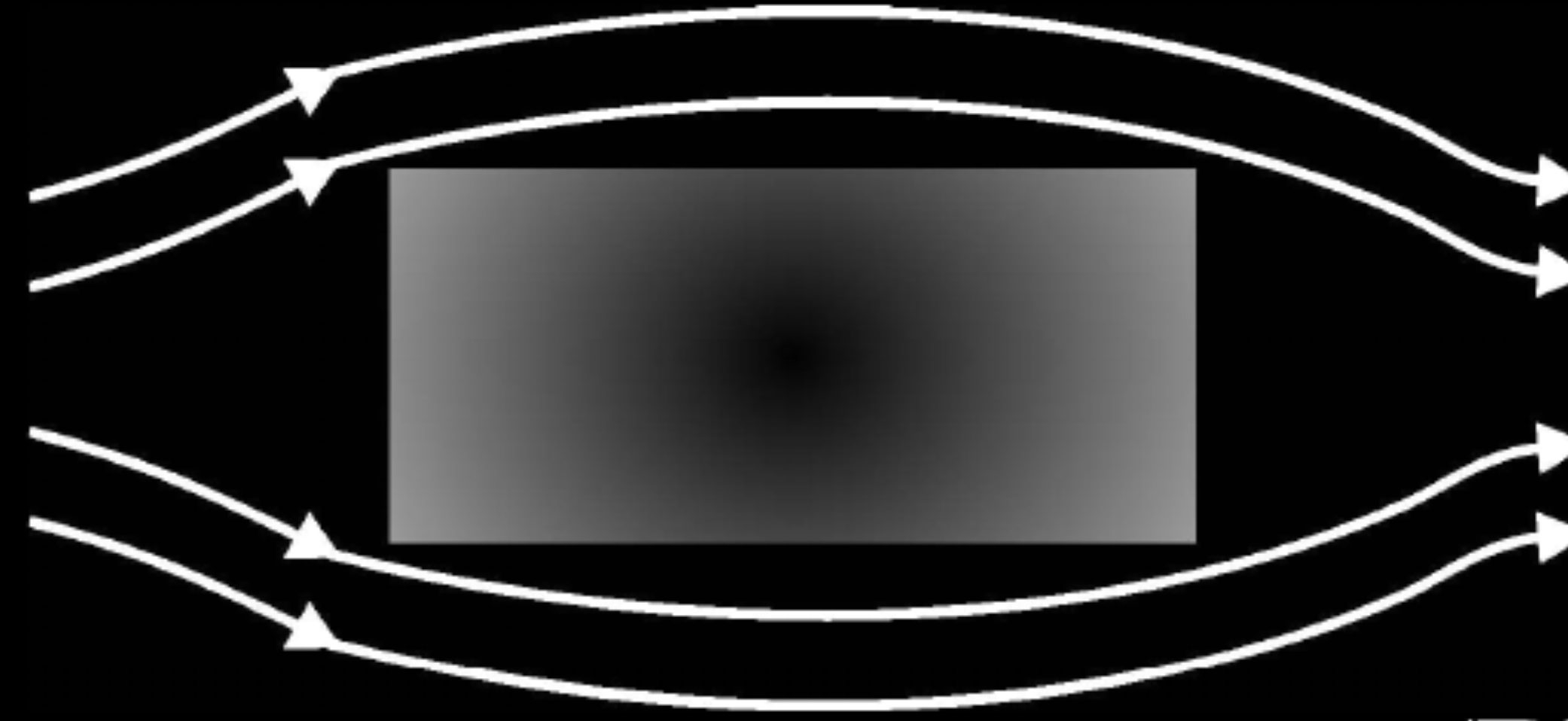
(C) Sodium Chloride

(B) Silicon

(D) Copper

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1



(A) Nickel

(C) Sodium Chloride

(B) Silicon

(D) Copper

(A) Nickel

3. Which of the following has its permeability less than that of free space ? **1**
- (A) Copper (B) Aluminium
- (C) Copper chloride (D) Nickel

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- (A) Copper (B) Aluminium
(C) Copper chloride (D) Nickel

(A) Copper

3. A bar magnet of pole strength (m) and magnetic moment (M) is cut perpendicular to its axis in two equal halves. The new pole strength (m') and magnetic moment (M') of each part are respectively :

- | | |
|----------------------------|----------------------------|
| (a) m and M | (b) m and $\frac{M}{2}$ |
| (c) $\frac{m}{2}$ and $2M$ | (d) $2m$ and $\frac{M}{2}$ |

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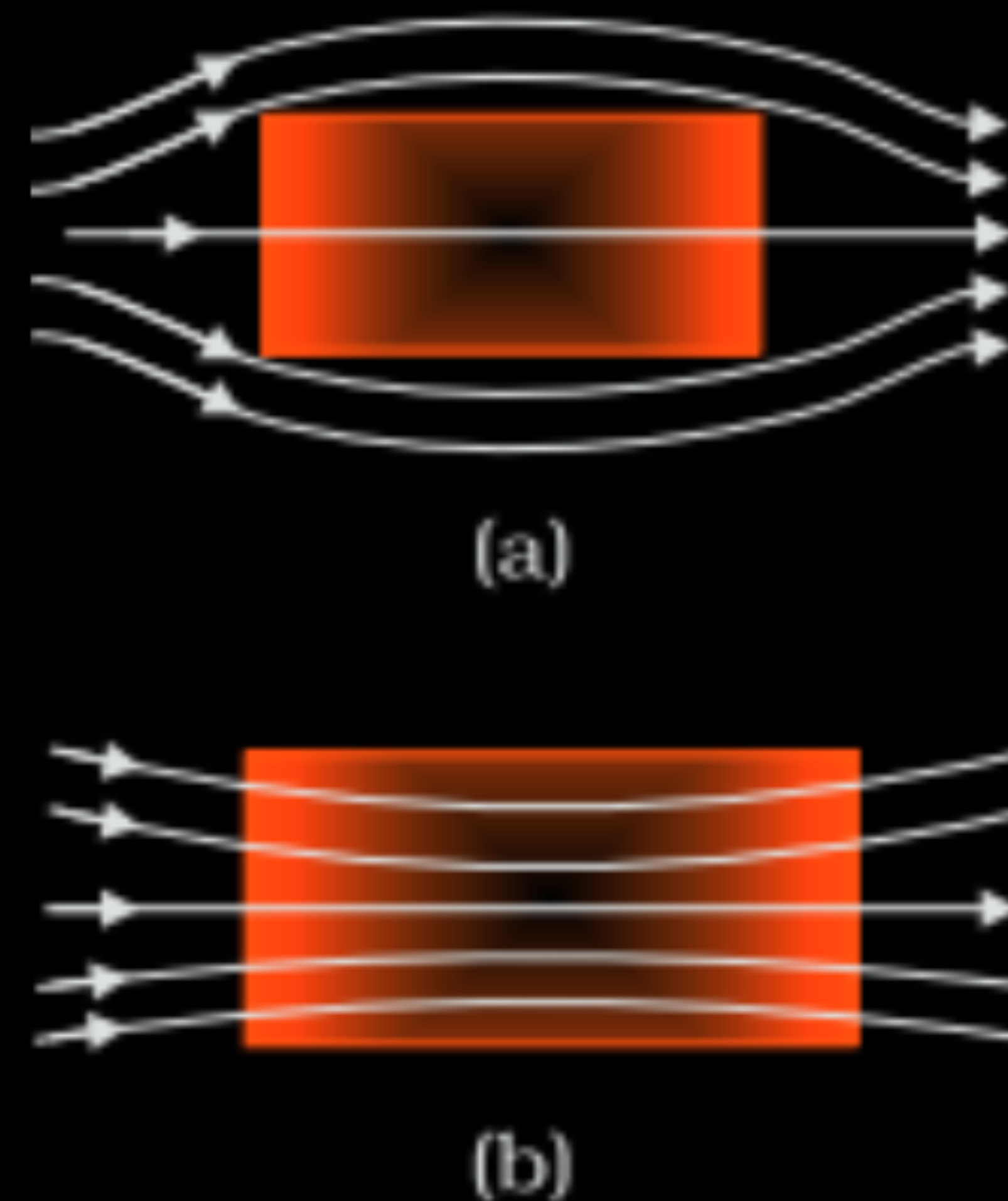
- | | |
|----------------------------|----------------------------|
| (a) m and M | (b) m and $\frac{M}{2}$ |
| (c) $\frac{m}{2}$ and $2M$ | (d) $2m$ and $\frac{M}{2}$ |

(b) m and $M/2$

- 20.** Two identical bars, one of a paramagnetic material and another of a diamagnetic material are kept in a uniform magnetic field. Show diagrammatically the modifications in the pattern of magnetic field in each case. How are the two materials affected by increase in temperature ?

20. Two identical bars, one of a paramagnetic material and another of a diamagnetic material are kept in a uniform magnetic field. Show diagrammatically the modifications in the pattern of magnetic field in each case. How are the two materials affected by increase in temperature?

2



$$\frac{1}{2} + \frac{1}{2}$$

No effect in case of diamagnetic materials.
Magnetization/ magnetic susceptibility decrease in case of paramagnetic materials.

$$\frac{1}{2}$$

$$\frac{1}{2}$$

3. A small bar, when placed near a magnet is repelled by it. This is because the bar is made of :

(a) Iron

(b) Copper

(c) Aluminium

(d) Nickel

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- 3.** A bar magnet is cut into two equal halves parallel to its magnetic axis. The physical quantity that remains unchanged is :
- | | |
|-----------------------|--------------------------------|
| (a) pole strength | (b) magnitude of magnetisation |
| (c) moment of inertia | (d) magnetic moment |

3. A bar magnet is cut into two equal halves parallel to its magnetic axis. The physical quantity that remains unchanged is :

- | | |
|-----------------------|--------------------------------|
| (a) pole strength | (b) magnitude of magnetisation |
| (c) moment of inertia | (d) magnetic moment |

(b) magnitude of magnetization

- (a) Show that a current carrying solenoid behaves like a small bar magnet. Obtain the expression for the magnetic field at an external point lying on its axis.

- (a) Show that a current carrying solenoid behaves like a small bar magnet. Obtain the expression for the magnetic field at an external point lying on its axis.

(a) A solenoid may be regarded as a combination of large number of identical circular current loops in which each behaves like a magnetic dipole. Hence, the current carrying solenoid will behave like a small bar magnet.

- (b) A bar magnet of dipole moment 3 Am^2 rests with its centre on a frictionless pivot. A force F is applied at right angles to the axis of the magnet, 10 cm from the pivot. It is observed that an external magnetic field of 0.25 T is required to hold the magnet in equilibrium at an angle of 30° with the field.
Calculate the value of F .
How will the equilibrium be effected if F is withdrawn ?

- (b) A bar magnet of dipole moment 3 Am^2 rests with its centre on a frictionless pivot. A force F is applied at right angles to the axis of the magnet, 10 cm from the pivot. It is observed that an external magnetic field of 0.25 T is required to hold the magnet in equilibrium at an angle of 30° with the field.
Calculate the value of F .
How will the equilibrium be effected if F is withdrawn ?

(b) In equilibrium

Restoring Torque = Deflecting Torque

$$F \times r = m B \sin \theta$$

$$F \times 10 \times 10^{-2} = 3 \times 0.25 \times \sin 30^\circ$$

$$F = \frac{3 \times 0.25 \times 1}{10 \times 10^{-2} \times 2}$$

$$= 3.75 \text{ N}$$

The magnet oscillates for sometime but finally aligns along the original direction of the external magnetic field.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

4. Above Curie temperature, a

1

- (A) ferromagnetic material becomes diamagnetic.
- (B) ferromagnetic material becomes paramagnetic.
- (C) paramagnetic material becomes ferromagnetic.
- (D) paramagnetic material becomes diamagnetic.

4. Above Curie temperature, a

1

- (A) ferromagnetic material becomes diamagnetic.
- (B) ferromagnetic material becomes paramagnetic.
- (C) paramagnetic material becomes ferromagnetic.
- (D) paramagnetic material becomes diamagnetic.

(B) ferromagnetic material becomes paramagnetic

CBSE 2019

Write one important property of a paramagnetic material.

OR

Do the diamagnetic substances have resultant magnetic moment in an atom in the absence of external magnetic field ?

Write one important property of a paramagnetic material.

OR

Do the diamagnetic substances have resultant magnetic moment in an atom in the absence of external magnetic field ?

Any one property of paramagnetic materials.
(e.g.
(i) It attracts field lines, weakly.
(ii) It moves from weaker towards stronger field. or any
other property.)

OR

No.

1

1

- (a) Show that the time period (T) of oscillations of a freely suspended magnetic dipole of magnetic moment (m) in a uniform magnetic field (B) is given by $T = 2\pi\sqrt{\frac{I}{mB}}$, where I is a moment of inertia of the magnetic dipole.
- (b) Identify the following magnetic materials :
- (i) A material having susceptibility $(\chi_m) = -0.00015$.
 - (ii) A material having susceptibility $(\chi_m) = 10^{-5}$.

a) Restoring torque $\tau = -mB \sin\phi = -mB \phi$ (for small ϕ) $\tau = I \frac{d^2\phi}{dt^2} = -mB\phi$ $\frac{d^2\phi}{dt^2} = \frac{-mB}{I} \phi$ Comparing with equation of angular SHM $\frac{d^2\phi}{dt^2} = -\omega^2\phi$ $\omega = \sqrt{\frac{mB}{I}}$ $T = 2\pi \sqrt{\frac{I}{mB}}$	$\frac{1}{2}$
b) i) diamagnetic	1
ii) Para magnetic	1

- (a) State Gauss's law for magnetism. Explain its significance.
- (b) Write the four important properties of the magnetic field lines due to a bar magnet.

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- (b) Write the four important properties of the magnetic field lines due to a bar magnet.

(a) Gauss's law for magnetism states that "The total flux of the magnetic field, through any closed surface, is always zero.

½

Alternatively

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

This law implies that magnetic monopoles do not exist" / magnetic field lines form closed loops

½

[Note: Award this 1 mark if the student just attempts it]

(b) Four properties of magnetic field lines

½

(i) Magnetic field lines always form continuous closed loops.

(ii) The tangent to the magnetic field line at a given point represents the direction of the net magnetic field at that point.

½

(iii) The larger the number of field lines crossing per unit area, the stronger is the magnitude of the magnetic field.

½

(iv) Magnetic field lines do not intersect.

½

Write three points of differences between para-, dia- and ferro- magnetic materials, giving one example for each.

Write three points of differences between para-, dia- and ferro- magnetic materials, giving one example for each.

	Diamagnetic	Paramagnetic	Ferromagnetic
1	$-1 \leq \chi < 0$	$-0 < \chi < \varepsilon$	$\chi \gg 1$
2	$0 \leq \mu_r < 1$	$1 \leq \mu_r < 1 + \varepsilon$	$\mu_r \gg 1$
3	$\mu < \mu_0$	$\mu > \mu_0$	$\mu \gg \mu_0$

Where ε is any positive constant.

[Note: Give full credit of this part if student write any other three correct difference]

Examples (Any one example of each type)

Diamagnetic materials: Bi, Cu, Pb, Si, water, NaCl, Nitrogen (at STP)

Paramagnetic materials: Al, Na, Ca, Oxygen (at STP), Copper chloride

Ferromagnetic materials: Fe, Ni, Co, AlNiCo

1/2

1/2

1/2

1/2

1/2

1/2

The magnetic susceptibility of magnesium at 300 K is 1.2×10^5 . At what temperature will its magnetic susceptibility become 1.44×10^5 ?

The magnetic susceptibility of magnesium at 300 K is 1.2×10^{-5} . At what temperature will its magnetic susceptibility become 1.44×10^{-5} ?

$$\chi_m \propto \frac{1}{T}$$

$$T_2 = \frac{\chi_{m1}}{\chi_{m2}} \times T_1$$

$$T_2 = \frac{1.2 \times 10^{-5}}{1.44 \times 10^{-5}} \times 300 = 250 K$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

The magnetic susceptibility χ of a given material is -0.5 . Identify the magnetic material.

The magnetic susceptibility χ of a given material is -0.5 . Identify the magnetic material.

It is diamagnetic material

1

- (a) Define the term magnetic susceptibility and write its relation in terms of relative magnetic permeability.
- (b) Two magnetic materials A and B have relative magnetic permeabilities of 0.96 and 500. Identify the magnetic materials A and B.

Write two properties of a material suitable for making (a) a permanent magnet, and (b) an electromagnet.

Write two properties of a material suitable for making (a) a permanent magnet, and (b) an electromagnet.

a) For making permanent magnet: (i) High retentivity (ii) High coercitivity (iii) High permeability (Any two) b) For making electromagnet: (i) High permeability (ii) Low retentivity (iii) Low coercivity (Any two)	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$
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- (i) Mention two properties of soft iron due to which it is preferred for making an electromagnet.
- (ii) State Gauss's law in magnetism. How is it different from Gauss's law in electrostatics and why ?

OR

Derive an expression for the axial magnetic field of a finite solenoid of length $2l$ and radius r carrying current I . Under what condition does the field become equivalent to that produced by a bar magnet ?

(i) Low coercivity and high permeability

$\frac{1}{2} + \frac{1}{2}$

(ii) The net magnetic flux through any closed surface is zero/

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$$\oint B \cdot ds = 0$$

$\oint E \cdot ds = \frac{q}{\epsilon_0}$ / The net electric flux through any closed surface is $\frac{1}{\epsilon_0}$

$\frac{1}{2}$

times the net charge.

$\frac{1}{2}$

which indicates magnetic monopoles do not exist/ magnetic poles

always exists in pairs

[Note : If the student just states Gauss's Law in electrostatics these 2 marks may be awarded.]

$$\text{Thus } B = \frac{\mu_0 2m}{4\pi r^3}$$

$\frac{1}{2}$

This is also the far axial magnetic field of a bar magnet. Hence, the magnetic field, due to current carrying solenoid along its axial line is similar to that of a bar magnet for far off axial points.

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In what way is the behaviour of a diamagnetic material different from that of a paramagnetic, when kept in an external magnetic field ?

In what way is the behaviour of a diamagnetic material different from that of a paramagnetic, when kept in an external magnetic field ?

A diamagnetic specimen would move towards the weaker region of the field while a paramagnetic specimen would move towards the stronger region./ A diamagnetic specimen is repelled by a magnet while a paramagnetic specimen moves towards the magnet./ The paramagnetic get aligned along B and the diamagnetic perpendicular to the field.	1
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